Mediacrest Training College

AI and Machine Learning Foundation Program

Week 1 Comprehensive Assessment

Total Marks: 10 — Duration: Self-Paced — Due: 10th October 2026

Instructions

This comprehensive assignment evaluates your understanding of Week 1 concepts, including the AI Family Tree, Data & Ethics, the Kenyan AI Context, and Prompt Engineering.

- Answer all questions in the spaces provided (or in your submitted document).
- Be specific and use local Kenyan examples where applicable.
- **Submission Format:** A single PDF document.
- **Naming Convention:** `FirstName_LastName_Week1_Assignment.pdf`
- **Submission Platform:** Mediacrest Student Portal / LMS.

Part 1: AI Concepts & The Paradigm Shift (2 Marks)

Context: Traditional programming relies on explicit, human-written rules, whereas Machine Learning (ML) relies on data to find patterns.

1. **The Paradigm Shift (1 Mark):** Briefly explain the fundamental difference in *inputs* and *outputs* between Traditional Programming and Machine Learning.

2. **Application to Local Context (1 Mark):** Imagine you are designing a system to detect fraudulent M-Pesa transactions. Provide **one** example of a *hard-coded rule* a traditional programmer might write, and **one** example of the *type of data* an ML model would use to learn fraud patterns.

Part 2: AI Ethics, Bias & Local Context (3 Marks)

Context: "Garbage In, Garbage Out" (GIGO) and data bias can lead to severe real-world consequences, especially when models trained on Western data are applied to the African context.

1. **Understanding Bias (1.5 Marks):** Explain how *Historical Data* and *Proxy Variables* can cause an AI credit-scoring model to unfairly reject a *mama mboga* (vegetable vendor) who has no formal payslip but excellent daily M-Pesa transaction records.

2. **Kenyan AI Applications (1.5 Marks):** Choose **one** of the following local AI use cases: *(A) AgriTech (Crop Disease Detection)* OR *(B) FinTech (Mobile Lending).* Briefly describe how the AI works and list **two** specific data inputs it requires to function effectively in Kenya.

Part 3: Prompt Engineering & Tool Application (3 Marks)

Context: Generative AI tools (like ChatGPT or Gemini) require structured communication to yield high-quality results. You must use the **R-C-I-F** framework: **R**ole, **C**ontext, **I**nstruction, **F**ormat.

1. **Prompt Construction (3 Marks):** You are a customer support manager for a local Kenyan tour operator. A client is angry because their safari transport was delayed by 3 hours due to heavy rains in the Maasai Mara.
   Task: Write a **single, comprehensive prompt** that you would feed into ChatGPT to generate an apology email. Your prompt **must** explicitly contain and label the following four components:

   - **Role:** (Who is the AI acting as?)
   - **Context:** (What is the background situation?)
   - **Instruction:** (What exactly should the AI write?)
   - **Format/Constraints:** (Length, tone, language, e.g., include a Swahili phrase).

Write your R-C-I-F Prompt here:

Part 4: "Invisible AI" in Daily Life (2 Marks)

Context: AI is not just conversational bots; it operates quietly behind the scenes in the apps we use daily.

1. **AI Audit (2 Marks):** Identify **one** "invisible" AI tool or algorithm you interact with weekly (e.g., TikTok's For You Page, Gmail Spam Filter, Google Maps routing).
    - **Name of Tool/Feature:**
    - **Specific Task it Performs:**
    - **Data Hypothesis:** What specific data do you hypothesize the AI is analyzing to perform this task?

Grading Rubric (For Instructor Use)

"With great computational power comes great responsibility." — End of Assignment —

Section	Marks	Criteria for Full Marks
Part 1: Concepts	2	Clearly distinguishes Inputs/Outputs of ML vs Traditional. Provides logical M-Pesa rule vs data example.
Part 2: Ethics & Context	3	Accurately explains GIGO/Bias regarding informal sector. Correctly identifies local AgriTech/FinTech use case and data inputs.
Part 3: Prompting	3	Prompt contains all 4 R-C-I-F elements. Context is highly specific to the Kenyan scenario. Constraints are measurable.
Part 4: Invisible AI	2	Identifies a non-generative AI tool. Provides a logical, well-reasoned hypothesis on the underlying data used.
TOTAL	**10**	